Appendix E: Acute Treatment Algorithm

Part 1: Diagnostic Algorithm

≥ 20 weeks pregnant **OR** pregnant in last 6 weeks?

YES

***Presenting Symptoms**

- Headache, visual complaints (most common precursor to eclampsia)
- Altered mental status, seizure, CVA
- Abdominal pain-especially RUQ, epigastric pain
- SOB, pulmonary edema
- Oliguria

If any of these are present with no other etiology, preeclampsia with severe features is suspected and magnesium sulfate should be considered.

YES

First: MEASURE BP then SEND LABS

CBC, AST, ALT, LDH, serum creatinine, urine protein, urine analysis, uric acid (optional)

SBP ≥ 160 / DBP ≥ 110
HYPERTENSIVE EMERGENCY

Repeat BP in 15 minutes
If sustained ≥ 160/ ≥ 110

OB Evaluation
IMMEDIATE

Initiate antihypertensives

SBP 140-159 / DBP 90-109
HYPERTENSION

OB Evaluation
Within 60 minutes

Serial BP q15min

IF BP INCREASES TO
SBP ≥ 160 OR DBP ≥ 110
Initiate antihypertensives

Notify provider if patient condition changes

SBP < 140 / DBP < 90
NORMAL

OB Evaluation
Within 60 minutes

Serial BP q15min

Patients with symptoms have preeclampsia with severe features despite initial 'normal BP'

IF BP INCREASES TO
SBP ≥ 160 OR DBP ≥ 110
Initiate antihypertensives

Notify provider if patient condition changes

Preeclampsia with severe features:

- SBP ≥160 mm Hg or DBP ≥ 110 mm Hg on 2 occasions at least 4 hours apart (unless antihypertensive therapy is initiated before this time)
- Thrombocytopenia
- Impaired liver function that is not accounted for by alternative diagnoses indicated by abnormally elevated liver enzymes or by severe persistent right upper quadrant or epigastric pain
- Renal insufficiency
- Pulmonary edema
- New-onset headache unresponsive to medication and not accounted for by alternative diagnoses
- Visual disturbances

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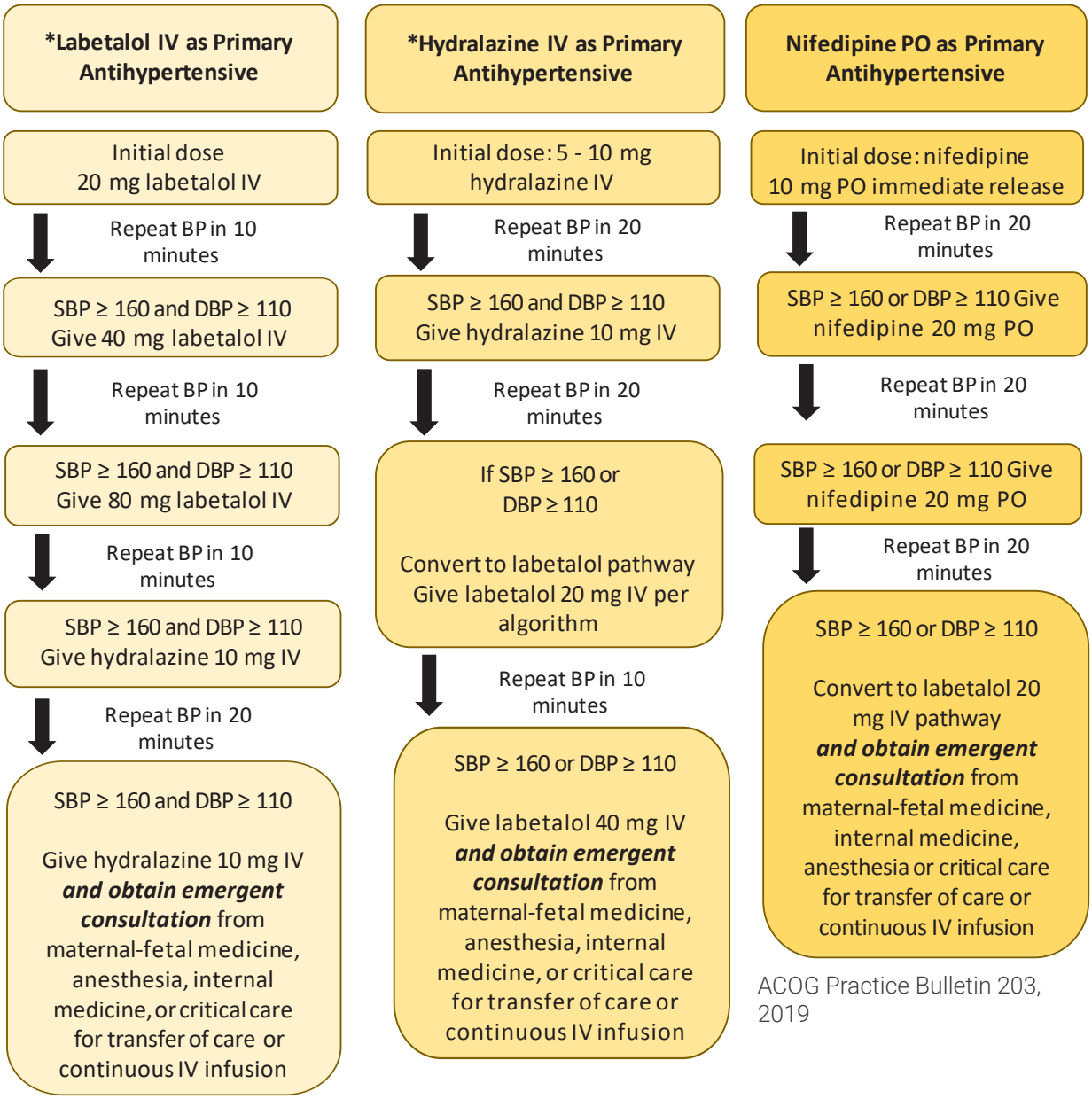
This figure was adapted from the Improving Health Care Response to Preeclampsia: A California Quality Improvement Toolkit, funded by the California Department of Public Health, 2014; supported by Title V funds.

Part 2: Antihypertensive Treatment Algorithm for Hypertensive Emergencies

Treatment Recommendations for Sustained Systolic BP ≥ 160 mm Hg or Diastolic BP ≥ 110 mm Hg

*Antihypertensive treatment and magnesium sulfate should be administered simultaneously. If concurrent administration is not possible, antihypertensive treatment should be 1st priority.



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Target BP: 130-150/80-100 mm Hg

Once BP threshold is achieved:

- Q10 min for 1 hr
- Q15 min for 1 hr
- Q30 min for 1 hr
- Q1hr for 4 hrs

*Intravenous hydralazine or labetalol should be given over 2 minutes. In the presence of sinus bradycardia or a history of asthma, hydralazine or nifedipine are preferred as initial agents. If maternal HR > 110, labetalol is preferred.

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Part 3: Magnesium Dosing and Treatment Algorithm for Refractory Seizures

Magnesium: Initial Treatment

1. Loading Dose: 4-6 gm over 20-30 minutes (6 gm for BMI > 35)
2. Maintenance Dose: 1-2 gm per hour
3. Close observation for signs of toxicity
   - Disappearance of deep tendon reflexes
   - Decreased RR, shallow respirations, shortness of breath
   - Heart block, chest pain
   - Pulmonary edema
4. Calcium gluconate or calcium chloride should be readily available for treatment of toxicity

For recurrent seizures while on magnesium

1. Secure airway and maintain oxygenation
2. Give 2nd loading dose of 2-4 gm Magnesium over 5 minutes
3. If patient still seizing 20 minutes after 2nd magnesium bolus, consider one of the following:
   - Midazolam 1-2 mg IV; may repeat in 5-10 min
     OR
   - Diazepam 5-10 mg IV slowly; may repeat q15 min to max of 30 mg
     OR
   - Phenytoin 1,250 mg IV at a rate of 50 mg/min
   - Other medications have been used with the assistance of anesthesia providers such as:
     - Sodium thiopental
     - Sodium amobarbital
     - Propofol
4. Notify anesthesia
5. Notify neurology and consider head imaging

Seizures Resolve

1. Maintain airway and oxygenation
2. Monitor vital signs, cardiac rhythm/EKG for signs of medication toxicity
3. Consider brain imaging for:
   - Head trauma
   - Focal seizure
   - Focal neurologic findings
   - Other suspected neurologic diagnosis
4. Reassure patient with information, support
5. Debrief with team before shift end

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